



International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
B. Sc. in CSE, Class Test Examination, Spring-2024
Course Code: MATH-1107, Course Title: Mathematics-I
Time: 20 min Marks: 10

Section 1BM

(Answer any two (02) questions. Figures in the right margin indicates full marks)

	Marks	CLO	DL
1. If $u = \frac{1}{r}$ and $r^2 = x^2 + y^2 + z^2$ then show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$	5	CLO2	C2
2. Evaluate the integral $\int \frac{dx}{(2x+1)\sqrt{4x+3}}$	5	CLO2	C2
3. Find the arc length of the curve $y = \frac{2}{3}x^{\frac{3}{2}}$ from $x = 3$ to $x = 8$ about the x-axis.	5	CLO3	C3



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